

Lines, Tables, and Big Jumps

ELA + MATH · WORD STUDY

Mr. H Math · Day 22

WRITE YOUR NAME — one letter per box, straight and neat

Date: _____

Score: ____ / 10

8 picnic tables need 3 garbage cans. How many cans do **120 tables** need? You could draw groups until your pencil gives out — or you could find the one number that gets you from 8 to 120 in a single jump. Today you get both tools: the double number line for seeing it, and the ratio table for doing it fast.

DO NOW · SILENT START

3 min

Two patterns: **8, 16, 24, 32**, ____ and **3, 6, 9, 12**, _____. Fill in the next number in each, then write what is being added each step.

TURN & TALK

2 min

With your partner, stack the two patterns as pairs: (8, 3), (16, 6), (24, 9)... Say what stays the same about every pair even though the numbers keep growing.

Frames: "Every step adds _____ and _____." · "That is the same as multiplying both by _____."

Section A · Guess the Word

Fill in the blanks to find the number you multiply BOTH quantities by.

M U T I P I E R

The SAME missing letter goes in both dashed boxes. Write it here: _____

$8 \times \text{_____} = 120$, so the cans get multiplied by that same number.

Stepping one group at a time and multiplying both quantities are the same move — multiplying just does all the steps at once.

Section B · Meet the Two Tools

Use the word bank to complete each sentence.

double number line

ratio table

multiplier

equivalent

1 Two number lines stacked and scaled together make a _____.

Example: 8 and 3 line up; so do 16 and 6.

2 Rows or columns of equivalent ratios make a _____.

Example: 8 / 16 / 24 tables above 3 / 6 / 9 cans.

3 Numbers that line up vertically on a double number line are _____ ratios.

Example: 8 : 3, 16 : 6, 24 : 9 — all the same comparison.

4 The number you multiply both quantities by to jump ahead is the _____.

Example: $8 \times 15 = 120$, so $3 \times 15 = 45$.

Section C · Try It: Fill the Table

8 picnic tables for every 3 garbage cans. Study the jump in the worked example, then complete Marta's heartbeat table.

WORKED EXAMPLE

Picnic tables	8	16	24	120
Garbage cans	3	6	9	45

$8 \times 15 = 120 \rightarrow 3 \times 15 = 45$ cans

↑ one multiplier, both rows — no drawing needed.

YOUR TURN

Marta's heart beats **18 times every 15 seconds**. Complete the table, then explain the 180-beat column.

Seconds	15	30	45	
Heartbeats	18			180

What did you multiply by to reach 180 beats?

REFINE · SUMMARY — A double number line shows equivalent ratios one equal group at a time; a ratio table lets you skip the steps by multiplying — or dividing — **both** quantities by the same number. When the question jumps far ahead, hunt for the multiplier first. Next week: **graphing** a table of equivalent ratios, and using part-to-part and part-to-whole ratios together.

☆ Find the multiplier, skip the drawing! ☆

ELA + MATH

Word Study Meets Number Sense · Day 22 wksht · Prepares students for Graphing Equivalent Ratios (6.RP.A.3a)

Mr. H Math · Day 22 CCSS